Arabic Spoken Command Recognition Using MFCC and Machine Learning Techniques

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Abstract

Arabic automatic speech recognition remains challenging due to dialectal variation, phonetic complexity, and the limited availability of annotated datasets. This paper presents an isolated Arabic spoken command recognition system based on classical speech processing and machine learning techniques. A custom dataset consisting of 780 recordings was collected from 13 speakers, covering six Arabic commands with multiple repetitions. All recordings were preprocessed through resampling, silence removal, and amplitude normalization. Acoustic features were extracted using Mel-Frequency Cepstral Coefficients (MFCCs), and fixed-length feature vectors were formed using the mean and standard deviation of the coefficients.

Four classifiers k-Nearest Neighbors (KNN), Support Vector Machines (SVM), Random Forest, and Gaussian Mixture Models (GMM) were trained and evaluated. System performance was assessed using both random train-test split and leave-one-speaker-out (LOSO) cross-validation to evaluate speaker-independent generalization. Experimental results show that the SVM classifier achieved the best overall performance, reaching 100% accuracy under random split evaluation and the highest mean accuracy under LOSO evaluation. The findings demonstrate that MFCC-based representations combined with classical machine learning models provide an effective and robust solution for Arabic isolated command recognition.

1. Introduction

Automatic speech recognition (ASR) plays a central role in modern human, computer interaction systems, enabling intuitive and hands-free control of digital devices. Despite significant progress in speech recognition for widely spoken languages, Arabic speech recognition remains a challenging research problem due to dialectal variation, phonetic complexity, and the limited availability of publicly accessible datasets. These challenges are even more pronounced for command-based systems that rely on accurate recognition of short, isolated utterances.

This project focuses on the recognition of isolated Arabic spoken commands using classical speech processing and machine learning techniques. Such systems are particularly useful in applications such as voice-controlled robots, smart home devices, and assistive technologies, where a limited set of predefined commands must be recognized reliably and efficiently. Unlike large-vocabulary continuous speech recognition systems, isolated command recognition allows for simpler models while still highlighting core challenges in acoustic modeling and classification.

To address this problem, a custom Arabic spoken command dataset was created. The dataset consists of recordings from 13 speakers, including both male and female participants. Each speaker recorded six Arabic commands (**ابدأ** (start), **أغلق** (close), **افتح** (open), **توقف** (stop), **يمين** (right), and **يسار** (left)) with 10 repetitions per command, resulting in a total of 780 audio recordings. All recordings were collected in a controlled environment to minimize background noise and speaker interference.

The overall system follows a classical speech recognition pipeline. First, raw audio recordings are converted to a unified format and preprocessed through silence removal and amplitude normalization. Next, acoustic features are extracted using Mel-Frequency Cepstral Coefficients (MFCCs), which effectively capture the perceptual characteristics of speech signals. Finally, several machine learning classifiers, including k-Nearest Neighbors (KNN), Support Vector Machines (SVM), Random Forests, and Gaussian Mixture Models (GMM), are trained and evaluated to recognize the spoken commands.

The performance of the proposed system is evaluated using both random train–test splitting and leave-one-speaker-out (LOSO) cross-validation to assess speaker-independent generalization. Classification accuracy and confusion matrices are used as the primary evaluation metrics. Through this comparative analysis, the project aims to highlight the effectiveness of classical feature-based approaches for Arabic isolated command recognition and to identify the most suitable classifier for this task.

2. Literature Review

2.1 Arabic Speech Recognition Challenges

Arabic automatic speech recognition (ASR) systems face several linguistic and practical challenges that distinguish them from recognition systems developed for other languages. One major challenge is the coexistence of Modern Standard Arabic and multiple regional dialects, which differ significantly in pronunciation, vocabulary, and phonetic realization. These variations lead to high acoustic and lexical diversity, making it difficult to design models that generalize well across speakers and dialects. Additionally, the absence of standardized written forms for dialectal Arabic introduces further inconsistencies in speech data representation [1].

Another significant limitation in Arabic ASR research is the lack of large, high-quality annotated speech datasets, particularly for dialectal speech. Many existing datasets are limited in size, speaker diversity, or recording conditions, which negatively affects model robustness. As highlighted in recent studies, these constraints often motivate researchers to focus on constrained tasks such as isolated word or command recognition, where the vocabulary is limited and acoustic variability can be more effectively controlled [1].

2.2 Feature Extraction Using MFCC

Feature extraction is a fundamental component of speech recognition systems, as it determines how speech signals are represented for subsequent classification. Mel-Frequency Cepstral Coefficients (MFCCs) are among the most widely adopted features in speech processing due to their ability to capture perceptually relevant information from speech signals. MFCC extraction involves framing the speech signal, transforming it into the frequency domain, mapping frequencies onto the Mel scale, and applying a cepstral transformation to obtain a compact representation of the spectral envelope [2].

MFCC features have been extensively used in speech recognition because they provide a balance between computational efficiency and discriminative power. The literature emphasizes that MFCCs effectively represent speech characteristics while reducing redundancy in the signal. To further improve robustness and ensure fixed-length representations, statistical measures such as the mean and standard deviation of MFCC coefficients are commonly used. This approach facilitates the use of classical machine learning classifiers and has been shown to perform well in recognition tasks involving short and isolated speech signals [2].

2.3 Classification Approaches in Speech Recognition

After feature extraction, machine learning classifiers are used to map acoustic feature vectors to command labels. Support Vector Machines (SVMs) are supervised learning models that construct decision boundaries, known as hyperplanes, to separate data points from different classes with maximal margin. SVMs can employ kernel functions to transform input data into higher-dimensional spaces, enabling effective nonlinear classification while remaining resilient to noise in the training data [3].

Another widely used ensemble method is the Random Forest classifier, which combines predictions from multiple uncorrelated decision trees. Each tree in the forest is trained on a bootstrap sample of the data, and final class predictions are obtained through majority voting. By aggregating several weak learners, Random Forests reduce overfitting and improve generalization performance compared to single decision trees [4]. Figure 1 illustrates how Random Forest combines predictions from multiple decision trees using majority voting.

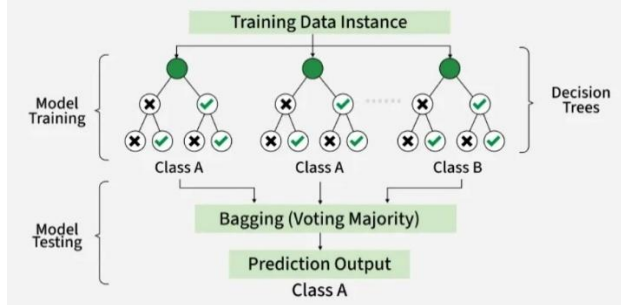

Figure 1. Random Forest classification using multiple decision trees and majority voting.

The k-Nearest Neighbors (KNN) algorithm is a non-parametric, instance-based learning method that assigns the class of a new sample based on the majority label among its nearest neighbors in the training set. The number of neighbors k and the distance metric (e.g., Euclidean) determine classification performance. KNN does not build an explicit model during training; instead, computation occurs at prediction time through distance comparisons [5].

Gaussian Mixture Models (GMMs) represent data as a weighted mixture of multiple Gaussian distributions. Although often associated with unsupervised clustering, GMMs can model class-conditional likelihoods by estimating parameters such as means and covariances of component Gaussians, allowing probabilistic classification based on the likelihood of feature vectors under each class model [6]. Figure 2 shows Gaussian components used in GMM to model the distribution of acoustic features.

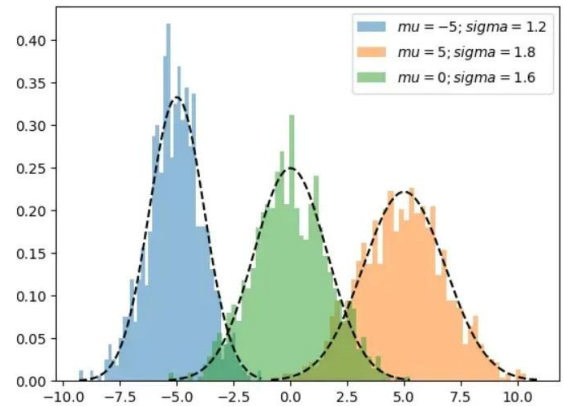

Figure 2. Example of Gaussian distributions used to model class-conditional feature distributions in a Gaussian Mixture Model.

3. Methodology (System Description)

3.1 System Overview

The proposed system follows a classical automatic speech recognition framework designed for isolated Arabic spoken command recognition. The system consists of five main stages: speech input, audio preprocessing, feature extraction, classification, and command recognition. Raw speech signals are first standardized through preprocessing techniques, after which discriminative acoustic features are extracted using Mel-Frequency Cepstral Coefficients (MFCCs). These features are then used to train and evaluate multiple machine learning classifiers. Figure 3 illustrates the overall architecture of the proposed system, starting from raw speech input and ending with the recognized Arabic command.

Figure 3. Block diagram of the proposed Arabic spoken command recognition system.

3.2 Dataset Description

A custom Arabic spoken command dataset was collected for this project. The dataset contains recordings from 13 speakers, including both male and female participants. Each speaker recorded six isolated Arabic commands: ابدأ (start), أغلق (close), افتح (open), توقف (stop), يمين (right), and يسار (left).

Each command was repeated 10 times per speaker, resulting in a total of 780 speech recordings. All recordings were collected in a quiet environment to reduce background noise and ensure consistency.

3.3 Audio Preprocessing

To improve recognition accuracy and ensure uniformity across recordings, several preprocessing steps were applied. First, all audio files were converted to mono WAV format and resampled to a sampling frequency of 16 kHz, which is commonly used in speech processing applications.

Silence removal was applied to eliminate non-speech segments at the beginning and end of each recording. In addition, amplitude normalization was performed to reduce variations in signal energy across different speakers and recording sessions. Signal normalization is defined as:

$$x_{norm}(n)=\frac{x(n)}{\max(|x(n)|)}$$

where x(n) represents the original speech signal and

$x_{norm}(n)$ is the normalized signal. These preprocessing steps help reduce unwanted variability and improve the robustness of feature extraction.

3.4 Feature Extraction Using MFCC

After preprocessing, Mel-Frequency Cepstral Coefficients (MFCCs) were extracted from each speech signal. MFCCs are widely used in speech recognition because they capture perceptually relevant spectral information and approximate the frequency resolution of the human auditory system.

The Mel frequency scale is defined as:

$$\mathrm{Mel}(f)=2595\log_{10}\left(1+\frac{f}{700}\right)$$

Each speech signal was segmented into short frames of 25 ms with an overlap of 10 ms between consecutive frames. This framing strategy allows the system to capture short-term spectral variations in speech while maintaining temporal continuity. For each frame, the Fast Fourier Transform (FFT) was applied to obtain the frequency spectrum. The spectrum was passed through a Mel filter bank, followed by a logarithmic operation and a Discrete Cosine Transform (DCT) to compute the MFCCs:

$$c_m=\sum_{k=1}^{K}\log(S_k)\cos\left[\frac{\pi m}{K}(k-0.5)\right]$$

where S_k represents the Mel filter bank energies and m denotes the MFCC index.

In this work, 13 MFCC coefficients were extracted per frame. To obtain fixed-length feature vectors suitable for machine learning classifiers, the mean and standard deviation of each coefficient were computed across all frames:

$$\mu_i=\frac{1}{N}\sum_{n=1}^{N}c_i(n)\ ,$$

$$\sigma_i=\sqrt{\frac{1}{N}\sum_{n=1}^{N}(c_i(n)-\mu_i)^2}$$

This resulted in a 26-dimensional feature vector for each speech recording.

3.5 Classification Models

Several machine learning classifiers were used to evaluate their effectiveness in recognizing Arabic spoken commands. These include k-Nearest Neighbors (KNN), Support Vector Machines (SVM), Random Forest, and Gaussian Mixture Models (GMM).

The SVM classifier aims to find an optimal separating hyperplane that maximizes the margin between different classes. The optimization problem is given by:

$$\min_{w,b}\frac{1}{2}\|w\|^2 \text{ subject to } y_i(w\cdot x_i+b)\geq 1$$

Random Forest is an ensemble learning method that combines multiple decision trees trained on random subsets of the data to improve generalization performance.

The KNN classifier assigns a class label based on the majority vote of the k closest feature vectors using a distance metric such as Euclidean distance.

For probabilistic modeling, GMMs were used to represent each command class as a mixture of Gaussian distributions:

$$p(x)=\sum_{k=1}^{k}\pi_k N(x\mid\mu_k,\Sigma_k)$$

Classification is performed by selecting the class that maximizes the likelihood of the observed feature vector.

3.6 Evaluation Strategy

System performance was evaluated using two validation strategies. First, a random train–test split was applied to measure overall classification accuracy. Second, a leave-one-speaker-out (LOSO) cross-validation scheme was used to evaluate speaker-independent performance.

Performance was primarily measured using classification accuracy, precision, recall, F1-score, and confusion matrices. To ensure reproducibility, all experiments were conducted using fixed random seeds and identical feature representations across all classifiers.

4. Experiments and Results

Part A: Random Split Evaluation

4.1 Experimental Setup

The experiments were conducted using the collected Arabic spoken command dataset described in Section 3. The dataset contains 780 audio recordings corresponding to 13 speakers, 6 commands, and 10 repetitions per command. After preprocessing and MFCC feature extraction, each recording was represented by a 26-dimensional feature vector.

For the random split evaluation, the dataset was divided into 80% training data and 20% testing data, resulting in 156 test samples (26 samples per command). This setup evaluates overall classification performance under standard conditions where training and test samples may include data from the same speakers.

System performance was evaluated using classification accuracy, precision, recall, F1-score, and confusion matrices to provide both quantitative and qualitative insights.

4.2 Quantitative Results (Random Split Evaluation)

Table 1 summarizes the classification accuracies obtained using the random split evaluation strategy.

Table 1:Classification Accuracy for Random Split Evaluation

Classifier	Accuracy (%)
KNN	98.08
Random Forest	96.79
SVM	100.00
GMM	98.72

The SVM achieved the highest performance with 100% accuracy, indicating perfect separation between all six command classes. KNN and GMM also demonstrated strong performance, exceeding 98% accuracy, while Random Forest achieved slightly lower accuracy.

4.3 Per-Class Performance Analysis

Detailed classification reports show that most commands achieved precision and recall values close to or equal to 1.00. Minor misclassifications were observed mainly for the commands أغلق (aghliq), ابدأ (ibda), and يسار (yasar) in the KNN, Random Forest, and GMM models. These errors are likely due to acoustic similarities between certain commands and speaker-dependent articulation differences.

The SVM classifier achieved perfect precision, recall, and F1-score for all commands.

4.4 Qualitative Results: Confusion Matrix Analysis

Confusion matrices provide a visual representation of classification behavior. The SVM confusion matrix shows no off-diagonal elements, confirming perfect classification. In contrast, KNN, Random Forest, and GMM exhibit a small number of off-diagonal entries corresponding to minor misclassifications.

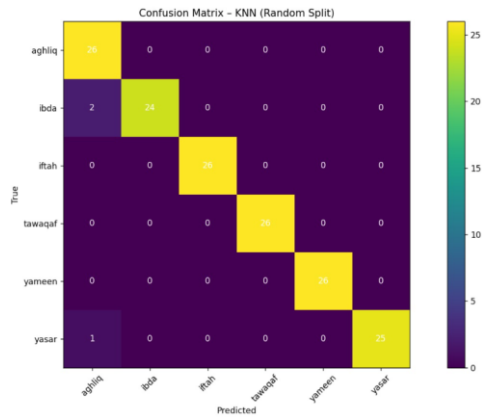

Figure 4: Confusion Matrix - KNN(Random Split)

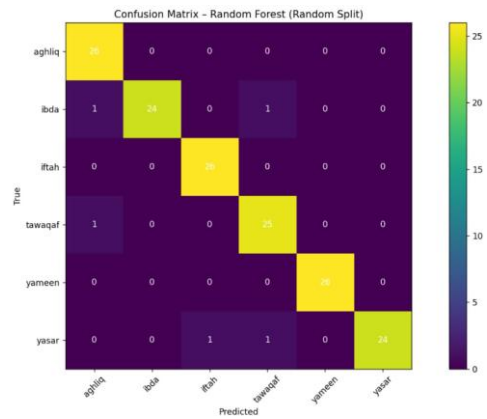

Figure 5: Confusion Matrix - Random Forest (Random Split)

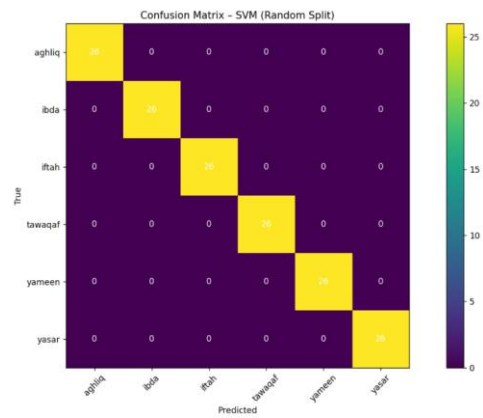

Figure 6:Confusion Matrix - SVM(Random Split)

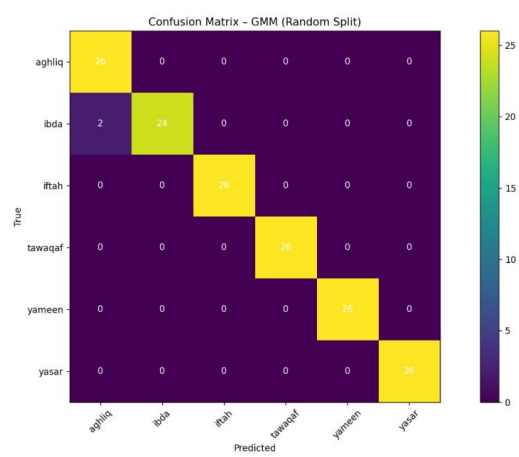

Figure 7: Confusion Matrix - GMM(Random Split)

Part B: Leave-One-Speaker-Out (LOSO) Evaluation

4.5 Experimental Setup

To evaluate speaker-independent generalization, a LOSO strategy was employed. In each fold, one speaker was used for testing and the remaining 12 for training. Final performance was computed as the mean and standard deviation across all folds.

4.6 Quantitative Results (LOSO Evaluation)

Table 2. LOSO Evaluation Accuracy (Mean ± Std)

Classifier	Accuracy (%)
KNN	82.96 ± 12.01
Random Forest	90.14 ± 6.90
SVM	91.30 ± 8.18
GMM	87.42 ± 6.05

SVM achieved the highest mean accuracy, followed by Random Forest. KNN showed the lowest accuracy and highest variance, indicating sensitivity to speaker variation.

4.7 Qualitative Results: Confusion Matrix Analysis

Figures 8- present the confusion matrices obtained from the Leave-One-Speaker-Out (LOSO) evaluation, summed across all folds for each classifier. The confusion matrices reveal that most classification errors occur between commands with similar phonetic structures, such as *aghlīq* and *tawaqaf*, particularly for the KNN and GMM classifiers.
The SVM and Random Forest classifiers demonstrate stronger diagonal dominance, indicating more consistent recognition performance across different speakers. Notably, the SVM classifier maintains high recognition rates for all commands, even under speaker-independent conditions, confirming its robustness when combined with MFCC-based features.

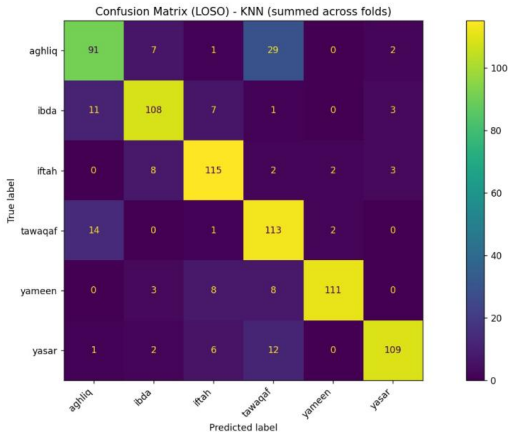

Figure 8: Confusion Matrix - KNN (LOSO)

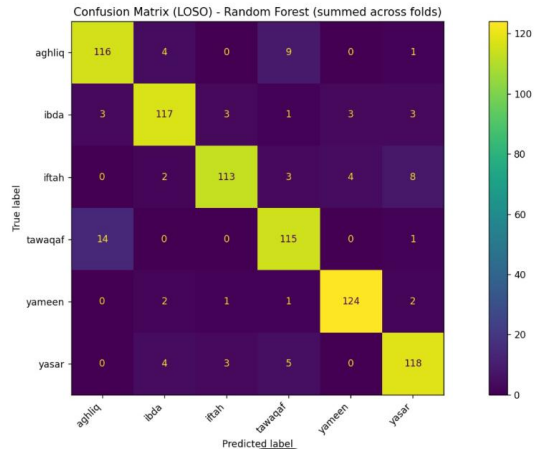

Figure 9: Confusion Matrix - Random Forest (LOSO)

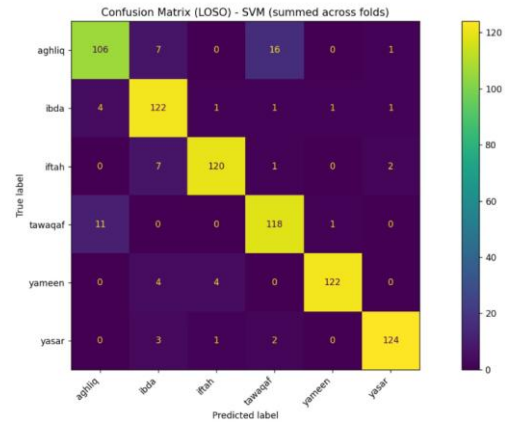

Figure 10: Confusion Matrix - SVM (LOSO)

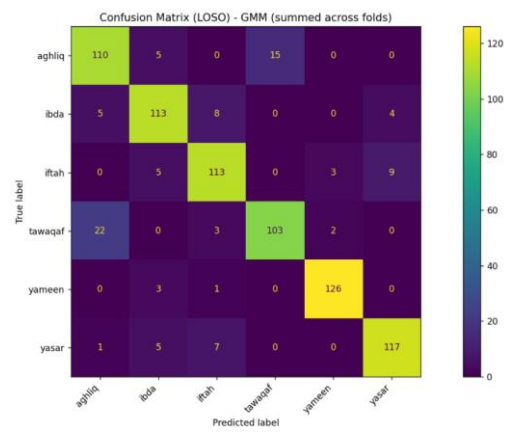

Figure 11: Confusion Matrix - GMM (LOSO)

4.8 Per-Class Performance Analysis

Under the Leave-One-Speaker-Out (LOSO) evaluation, per-class performance shows increased variability compared to the random split setting due to inter-speaker differences. Detailed classification reports indicate that most commands still achieved high precision and recall values; however, noticeable performance degradation was observed for certain commands when evaluated on unseen speakers.
Misclassifications occurred mainly for the commands *aghlīq*, *tawaqaf*, and *yasar*, particularly in the KNN and GMM classifiers. These errors are primarily attributed to phonetic similarities between some commands and to speaker-dependent variations in pronunciation and articulation.
The SVM and Random Forest classifiers demonstrated more stable per-class performance across speakers, achieving higher and more consistent precision and recall values for all six commands. In particular, the SVM classifier maintained strong discriminative capability under speaker-independent conditions, confirming its robustness when combined with MFCC-based feature representations.

4.8 Comparison Between Random Split and LOSO Evaluations

A clear performance gap is observed between random split and LOSO evaluations. While random split results reach up to 100% accuracy, LOSO results decrease due to inter-speaker variability. This confirms that random split evaluation may overestimate real-world performance, whereas LOSO provides a more realistic assessment.

5. Conclusion and Future Work

This paper presented a classical Arabic isolated spoken command recognition system using MFCC feature extraction and machine learning classifiers. A custom dataset was collected and used to evaluate KNN, Random Forest, SVM, and GMM models under both random split and speaker-independent LOSO evaluation strategies.

The experimental results demonstrate that Support Vector Machines consistently outperform other classifiers, achieving perfect accuracy under random split evaluation and the highest mean accuracy under LOSO evaluation. These results confirm that MFCC-based acoustic features combined with discriminative classifiers are well suited for small-scale Arabic spoken command recognition tasks. The LOSO evaluation further highlighted the importance of speaker-independent testing, as performance decreases under unseen-speaker conditions due to inter-speaker variability.

Future work will focus on expanding the dataset with more speakers and dialects, incorporating deep learning models such as convolutional or recurrent neural networks, and evaluating system robustness under noisy and real-world environments. These extensions aim to improve generalization and move toward practical deployment of Arabic voice-controlled systems.

6. References

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